

| Question |     |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks available |          |          |           |          |           |
|----------|-----|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|-----------|
|          |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | AO1             | AO2      | AO3      | Total     | Maths    | Prac      |
|          | (c) |  | <p>Jack:<br/>The metal is made from copper (1)</p> <p>Karen:<br/>The metal could be zinc, bronze or copper (1) <b>ecf</b> must be consistent with answer to (b)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |          | 2        | 2         |          | 2         |
|          | (d) |  | <p>Procedure – Jack may have: (award 2 marks max)</p> <ul style="list-style-type: none"> <li>• Taken repeat readings <u>of diameter</u> (1)</li> <li>• Measure mass more accurately (1)</li> <li>• Added weights in incremental steps / obtained <u>varying extensions</u> [with varying masses] / use a larger mass (1)</li> <li>• Use travelling microscope / vernier scale for extension (1)</li> </ul> <p>Don't accept different lengths</p> <p>Analysis – Jack may have: (award 1 mark max)</p> <ul style="list-style-type: none"> <li>• Draw a graph of force v extension or stress v strain [to find a mean value or gradient] (1)</li> </ul> |                 |          | 3        | 3         |          | 3         |
|          |     |  | <b>Question 4 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>4</b>        | <b>5</b> | <b>5</b> | <b>14</b> | <b>8</b> | <b>14</b> |